CHEST

Single-Night Diagnosis of Sleep Apnea Contributes to Inconsistent Cardiovascular Outcome Findings

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BACKGROUND: Single-night disease misclassification of OSA due to night-to-night variability may contribute to inconsistent findings in OSA trials.

RESEARCH QUESTION: Does multinight quantification of OSA severity provide more precise estimates of associations with incident hypertension?

STUDY DESIGN AND METHODS: A total of 3,831 participants without hypertension at baseline were included in simulation analyses. Included participants had ≥ 28 days of nightly apnea-hypopnea index (AHI) recordings via an under-mattress sensor and ≥ three separate BP measurements over a 3-month baseline period followed by ≥ three separate BP measurements 6 to 9 months postbaseline. Incident hypertension was defined as a mean systolic BP ≥ 140 mm Hg or a mean diastolic BP ≥ 90 mm Hg. Simulated trials (1,000) were performed, using bootstrap methods to investigate the effect of variable numbers of nights (x = 1-56 per participant) to quantify AHI and the ability to detect associations between OSA and incident hypertension via logistic regression adjusted for age, sex, and BMI.

RESULTS: Participants were middle-aged (mean ± SD, 52 ± 12 y), mostly male (91%), and overweight (BMI, 28 ± 5 kg/m^2). Single-night quantification of OSA failed to detect an association with hypertension risk in 42% of simulated trials ($\alpha = .05$). Conversely, 100% of trials detected an association when AHI was quantified over ≥ 28 nights. Point estimates of hypertension risk were also 50% higher and uncertainty was five times lower during multinight vs single-night simulation trials.

INTERPRETATION: Multinight monitoring of OSA allows for better estimates of hypertension risk and potentially other adverse health outcomes associated with OSA. These findings have important implications for clinical care and OSA trial design. CHEST 2023; 164(1):231-240

KEY WORDS: misdiagnosis; polysomnography; sleep-disordered breathing; wearables

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ABBREVIATIONS: AHI = apnea-hypopnea index; WSA = Withings sleep analyzer

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Take-home Points

Study Question: Does multinight quantification of OSA severity provide more precise estimates of associations with incident hypertension?

Results: Single-night quantification of OSA is less likely to detect an association with incident hypertension compared with multinight OSA severity estimates.

Interpretation: Night-to-night variability in OSA severity is a major source of variance that negatively impacts the ability to detect associations between OSA severity and health outcomes.

OSA is estimated to affect nearly 1 billion people worldwide[1] and is associated with poor health outcomes.[2-8] Several large studies and a meta-analysis have shown that there is considerable night-to-night variability in OSA severity in many patients, which can lead to misclassification[1,9,10] and that variation in OSA severity is associated with poorer cardiovascular outcomes.[11-13] The probability of misdiagnosis based on a single-night diagnostic study is estimated to be between 20% and 50%, with higher misdiagnosis probabilities for mild to moderate OSA.[1,9,10] Furthermore, estimation of the apnea-hypopnea index (AHI) using a single-night diagnostic test may differ by as much as 50% from the average severity based on a multinight diagnostic average.[14]

Night-to-night variability in OSA severity not only has important implications for clinical care and management of OSA,[15-17] but, as highlighted by Punjabi et al,[9] also raises questions regarding the validity of existing clinical trials that have historically relied on a single night of measurement to define OSA severity. Wide confidence limits around estimates of OSA severity may also contribute to underestimation of the overall health burden associated with OSA in epidemiologic studies.[7,8,18,19] Variability may also explain, at least in part, heterogeneity in clinical outcomes observed in treatment trials.[20-23] For example, an association between OSA and hypertension has been demonstrated in some cross-sectional and longitudinal trials,[5,7,18,24] but not others.[25,26] Similarly, CPAP treatment trials that have studied the effect of therapy on BP reduction have shown marginal improvement and at times inconsistent findings.[20,22,27-29] We hypothesized that, due to high night-to-night variability of OSA severity, single-night diagnosis may reduce the ability to detect associations with OSA and important health outcomes such as incident hypertension and diminish the magnitude of the effect. Therefore, this study sought to investigate the effect of single vs multinight assessment of OSA severity on associations with incident hypertension in a large dataset in which multinight assessments of AHI and repeat BP measures were performed.

Study Design and Methods

Participants and Equipment

Data used in this study were derived from 12,287 participants who registered to use both under-mattress sleep sensor technology (Withings sleep analyzer [WSA]) and a US Food and Drug Administration-registered, validated[30] Withings BP cuff monitor between July 1, 2020, and March 30, 2021. The WSA is an under-the-mattress sensor device that can unobtrusively estimate AHI with good agreement with in-laboratory polysomnography-derived AHI to classify moderate-to-severe and severe OSA (88% sensitivity and 88% specificity to detect ≥ 15 events/h sleep and 86% sensitivity and 91% specificity to detect ≥ 30 events/h sleep).[1,31] The estimated AHI also has minimal bias with in-laboratory polysomnography-derived AHI when the AHI is considered as a continuous variable.[31] The Withings BP monitor comes with an instruction booklet that outlines how to take a BP measurement. The user is instructed to (1) rest for at least 5 min before taking a measurement, (2) be seated in a comfortable position and in a quiet area, (3) not speak during the measurements, and (4) perform the measurement on their left arm.

Inclusion criteria for the current analyses were ≥ 28 days of nightly WSA recordings and at least three repeat BP measurements collected on separate occasions between July 2020 and October 2020 (defined as "baseline" data). In addition, to be included in the analyses, participants had to have at least three repeat BP measurements between January 2021 and March 2021 (the "follow-up" data). All first-time users of the WSA were also prompted to enter their age, sex, height, and weight, from which the BMI was calculated. All participants provided informed consent for their deidentified data to be used for research purposes. This study was approved by the Flinders University Human Research Ethics Committee (Project No. 4291).

Exposure Variable

OSA severity was calculated as the x-night average AHI over the baseline recording period (between 1 and 56 nights). OSA severity categories were also defined using commonly accepted cutoffs[32] (< 5, no OSA; ≥ 5 and < 15, mild; ≥ 15 and < 30, moderate; and ≥ 30 events/h, severe OSA).

Outcome Variable

Hypertension was defined as a mean systolic BP ≥ 140 mm Hg or a mean diastolic BP ≥ 90 mm Hg at baseline and follow-up[33] periods. All participants with hypertension in the baseline period were excluded. The primary outcome was incident hypertension at follow-up.

Statistical Analysis

The methodology used in this study is depicted in Figure 1. For each participant, an x-night average AHI was determined by

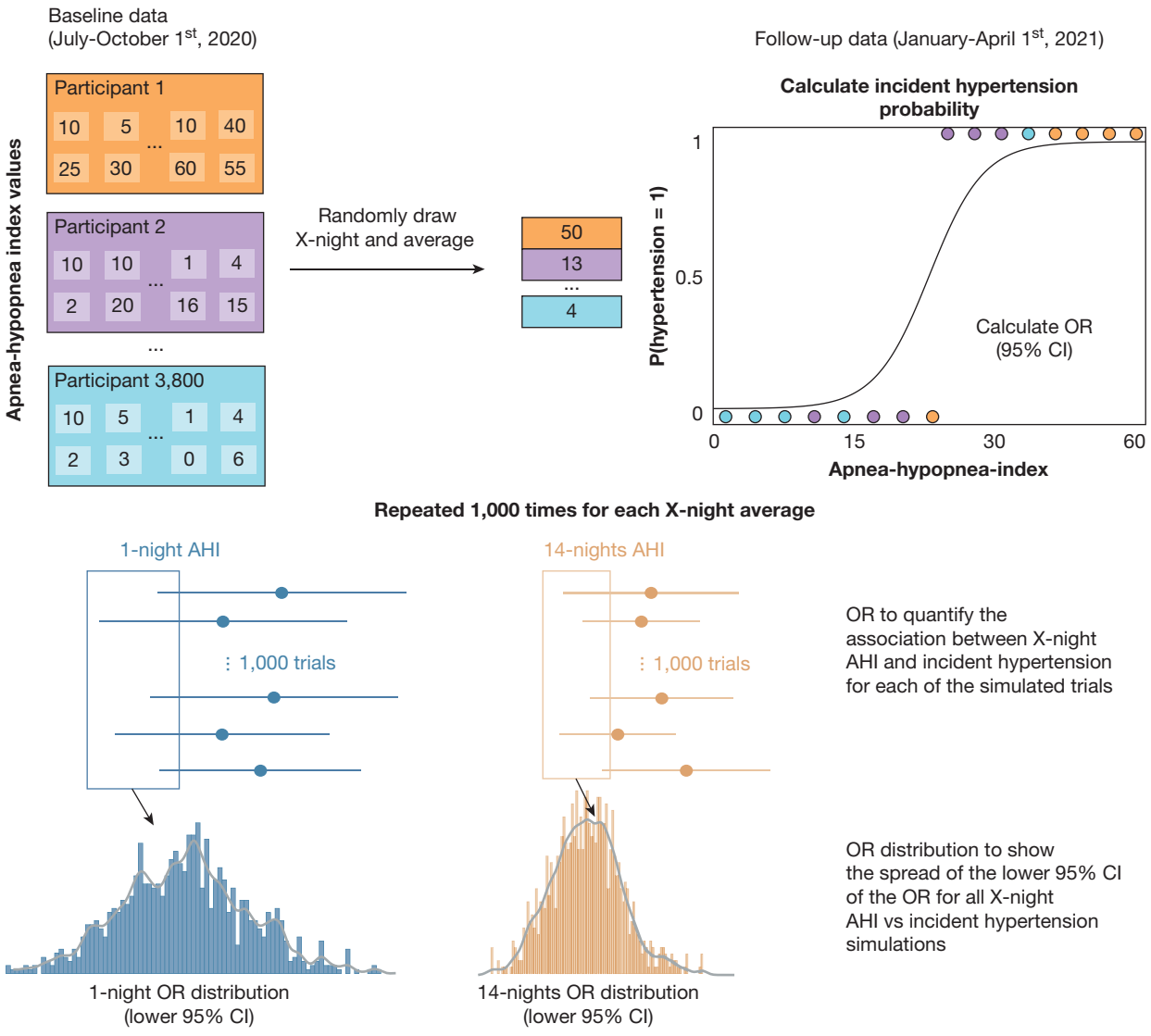


Figure 1 – *Bootstrap methodology used in this study. One thousand trials were created for each x-night apnea-hypopnea index (AHI) average by randomly selecting AHI values from the baseline period for each participant. The association between an x-night AHI and incident hypertension was determined by logistic regression, and the OR (95% CI) adjusted for age, BMI, and sex for each simulated trial was calculated. The distribution of the lower 95% CI of the OR for all the simulations was used to summarize the association between the x-night AHI and incident hypertension and to determine how many times the lower 95% CI crossed 1.*

using the mean AHI of *x* randomly selected nights from the 3-month baseline period. Sampling was repeated for each *x*-night average and for all participants 1,000 times. Thus, 1,000 dataset trial simulations for each *x*-night AHI average were performed to define OSA from which associations with incident hypertension were assessed. The optimal number of simulations was determined by convergence analysis. Consecutive days cannot be used to run 1,000 simulations consisting of up to 56 days of data per participant over a 3-month period. Accordingly, up to 56 nights of data within the 3-month period were randomly selected for each participant, and this procedure was repeated 1,000 times. We took the same approach for all other *x*-night AHI averages. For each simulation trial, the association between baseline AHI (continuous variable) and hypertension was determined by logistic regression controlled for age, sex, and BMI. Regression coefficients, ORs, and 95% CI were used to quantify hypertension risk for each 10-unit increase in AHI. This analysis was also repeated using clinical OSA severity categories.

Sensitivity Analyses

To further validate our findings and to account for any potential time-dependent effect in the association between OSA severity and incident hypertension, the analysis was repeated by additionally adjusting for the time between baseline and follow-up measurements.

Results

Participant Characteristics

A total of 4,675 participants had sufficient data to meet the initial baseline and follow-up monitoring inclusion criteria for the study. Of these participants, 844 (18.1%) had hypertension in the baseline period and were subsequently excluded. The remaining 3,831 participants were on average middle-aged, predominantly male, and overweight (Table 1). The prevalence of mild (64.4%) and moderate (30.3%) OSA was calculated on the basis of all available nights during the baseline period. Participants each had an average of 67 ± 17 nights of recording for a combined total exceeding 800,000 nights of data. The median number of BP entries for each participant during the baseline period was (median, interquartile range) 24 (9-64) and 16 (7-54) during the follow-up. There were 409 cases (10.2%) of incident hypertension during the follow-up period. The coefficient of variation for systolic and diastolic BP was similar at follow-up vs baseline measurement (coefficient of variation systolic BP, follow-up vs baseline: 6.5% ± 2.8% vs 6.6% ± 2.6%; coefficient of variation diastolic BP, follow-up vs baseline: 6.9% ± 3.0% vs 7.0% ± 2.8%).

Number of Simulations

The estimated mean of the OR distribution reached stability after a minimum of 500 simulations (e-Figs 1 and 2). Conservatively, we therefore elected to run 1,000 simulations per *x*-night investigation. Furthermore, we tested the sampling scheme with and without replacement, which revealed little difference between the two schemes (approximately 0.0005 for the mean OR).

Night-to-Night Variability and Its Impact on Associations Between OSA and Hypertension

The number of nights used to determine AHI had a direct impact on the association between AHI and incident hypertension (Fig 2). The number of simulated trials with a positive association between AHI and incident hypertension, using a 1-night AHI value, was 58%. In the remaining 42% of trials, the lower bound of the 95% CI was below 1. However, increasing the number of nights for AHI quantification reduced the proportion of nonsignificant trial findings (Fig 2). Seven and 14 nights of monitoring reduced the nonsignificant trial rate to 5% and 2%, respectively. Furthermore, an increased number of nights to quantify AHI reduced uncertainty regarding the association between AHI and hypertension as reflected by the distributions, along with their SDs and 95% CIs, which were much wider for 1-night (SD, 0.018) vs 14-night (SD, 0.006; three times lower) or 28-night AHI (SD, 0.004; 4.5 times lower). Each 10-point increase in 1-night AHI was associated with a 6.8% (bootstrapped 95% CI, 3.4%-10.3%) increase in hypertension risk whereas a 10-point increase in 56-night averaged AHI was associated with a 9.2% (8.5%-9.8%) increase in hypertension risk. Mean

TABLE 1] Baseline Characteristics of Population Sample

Variable	Category	Hypertension Status at Follow-up		
		All	Normotensive	Hypertensive
No.		3,831	3,422	409
Age, y		52 (12)	52 (12)	52 (12)
Sex	Male	3,496 (91%)	3,111 (91%)	385 (94%)
	Female	335 (9%)	311 (9%)	24 (6%)
BMI, kg/m^2		28 (5)	27 (5)	30 (5)
Baseline systolic BP, mm Hg		122 (9)	121 (9)	130 (7)
CV, %		6.6 (2.6)	6.6 (2.6)	6.8 (2.8)
Baseline diastolic BP, mm Hg		78 (7)	78 (6)	83 (5)
CV, %		7.0 (2.8)	7.0 (2.7)	7.2 (3.1)
Mean AHI (per h)		13 (14)	12 (13)	16 (16)
SD (per h)		6.2 (4.1)	6.1 (4.0)	7.3 (4.9)
OSA severity category	No	1,365 (36%)	1,245 (36%)	120 (29%)
	Mild	1,304 (34%)	1,179 (34%)	125 (31%)
	Moderate	776 (20%)	678 (20%)	98 (24%)
	Severe	386 (10%)	320 (10%)	66 (16%)

Average follow-up duration, 212 ± 25 d. Coefficient of variation is defined as (SD/mean) × 100. AHI = apnea-hypopnea index; CV = coefficient of variation.

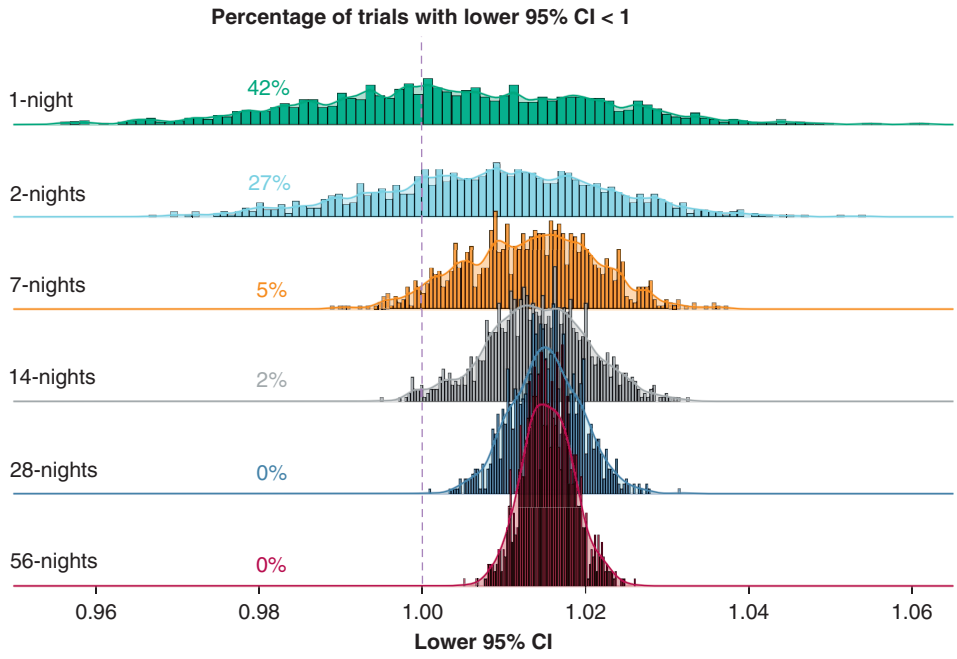


Figure 2 – *Distribution of the OR 95% CI lower bound from 1,000 simulations of the association between an x-night averaged AHI and incident hypertension. Percentages reflect the number of trials (in %) with a lower 95% CI below 1.*

point estimates and distributions are available in e-Figure 3. The increase in association certainty increased according to a logistic pattern, with rapid improvement evident with up to 10 to 15 nights of recording (e-Fig 4A), followed by more stable mean estimated OR with additional recording nights (e-Fig 4B).

Similar results were observed when OSA severity was categorized using standard clinical cutoffs (Fig 3). Severe OSA was nonsignificant in 37% of simulated trials based on a 1-night assessment, compared with 4% and 1% with 7 and 14 nights, respectively. Using a 56-night AHI, 98% of simulations did not show an increased

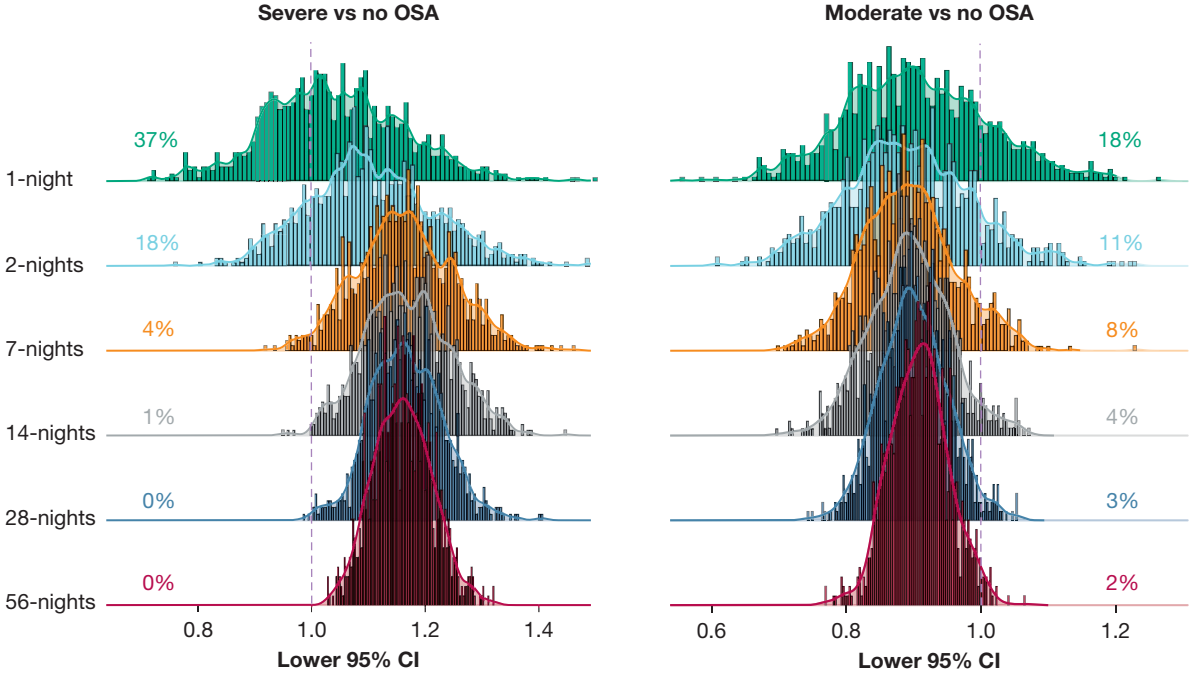


Figure 3 – *Distribution (1,000 simulations) of the OR 95% CI lower bound of the association between OSA severity categories (left, severe OSA; right, moderately severe OSA) determined using an x-night averaged AHI and incident hypertension.*

hypertension risk for moderately severe OSA vs no OSA. In contrast, a single-night AHI showed an increased hypertension risk in 18% of simulated trials for moderate OSA vs no OSA; likely to be attributable to type I error given that the 56-night AHI trials did not find an association.

Point estimates were also more variable for 1-night vs multinight AHI. For example, severe OSA classified from a 1-night AHI was associated with a 47% (bootstrapped 95% CI, 18%-81%) increase in hypertension risk vs no OSA. In contrast, severe OSA determined using a 56-night averaged AHI was associated with a 67% (51%-85%) increase in hypertension risk (e-Fig 5).

The uncertainty around the point estimates was approximately 2-fold greater in the 1-night average compared with the 28- or 56-night average AHI. The increase in certainty was lower for OSA severity categories (56 vs 1 night, 2-fold increase) compared with models using AHI as a continuous variable (56 vs 1 night, 4.5-fold increase), likely due to the intrinsic variability in the model and information loss through categorizing AHI.

Nonlinear Associations

The association between a 1-night AHI and incident hypertension was considerably more uncertain compared with a 14- or 56-night AHI. Findings are shown in Figure 4. The associations between an *x*-night AHI and incident hypertension were modeled with a cubic spline to allow for nonlinearity.[8,24]

Sensitivity Analyses

Further adjusting for the duration between baseline measures and outcomes measures did not change any of the main findings (e-Figs 6-8).

Discussion

With the use of approximately 9 months of nightly sleep apnea severity recordings with concurrent repeat daytime BP measures, we were able to run simulations to

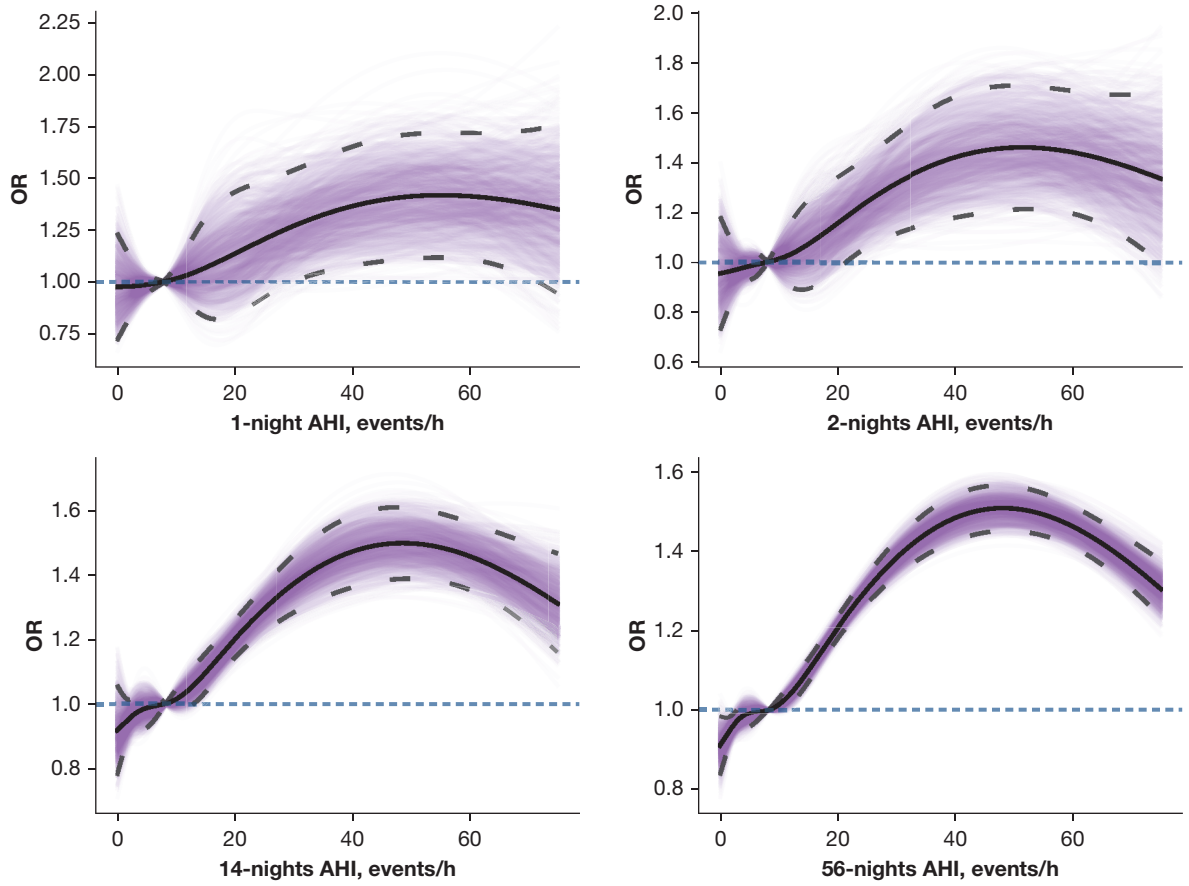


Figure 4 – *Association between an x-night apnea-hypopnea index (AHI) modeled using a restricted cubic spline and incident hypertension from 1,000 simulations. Each line represents a single trial. Note the much wider uncertainty for the 1-night AHI compared with the 14- or 56-night AHI. Associations are referenced to an AHI value of 8 (median of the population sample). Each purple line represents one trial, and black solid and dashed lines represent mean and 95% CI, respectively.*

investigate the effect of different combinations of variable nightly averages to quantify the AHI and the impact on incident hypertension risk detection. The current findings demonstrate that multinight measurement of OSA severity substantially increases the certainty regarding the association between OSA severity and incident hypertension risk. These results highlight the importance of consideration of night-to-night variability in understanding OSA disease risk profiles and in the design of future cohort studies and clinical trials.

The increased likelihood to detect significant associations between OSA and incident hypertension with additional nights of monitoring is striking and provides strong support that night-to-night variability in OSA severity contributes to inconclusive trial findings when limited to a single-night diagnosis. These results may also explain, at least in part, some of the heterogeneity in findings from treatment trials of OSA[20-23] and inconsistencies across epidemiologic trials between OSA severity and associations with important health outcomes.

In many ways, the field of hypertension/BP monitoring has several measurement insights directly relevant to OSA diagnosis, health impact assessment, and clinical management. For example, evidence indicates that multiple-measurement averaged data from home BP monitors are more strongly associated with cardiovascular events and mortality than office BP measurements.[34-36] Indeed, clinical guidelines for BP monitoring now suggest that patients with an office BP measurement close to clinical hypertension cutoffs should undertake multiple days of home BP monitoring (or ambulatory BP, if possible) to strengthen diagnostic confidence and to rule out/in white-coat and masked hypertension.[37] Similarly, limited diagnostic confidence in patients with mild/moderate sleep apnea from a single diagnostic study night who are most risk of being under- and overdiagnosed,[1,9] should also be considered for multinight OSA monitoring to help rule in/out similar measurement bias effects.

Conceptual Implications for Study Design and Reporting

Our results suggest that noisy AHI estimates can induce a higher likelihood of both type II errors (missing a "true" effect) and type I errors (a false discovery trial). In the absence of multinight monitoring, it may be useful to consider increasing the level of significance of the statistical test to reduce type I errors to help avoid situations in which clinical decisions are drawn from statistical significance. For example, a more conservative α significance of .01 when the AHI was quantified using a single overnight study reduced the number of potentially false positive trials for the association between moderately severe sleep apnea and incident hypertension to less than 5%. The use of a more conservative α significance level such as .01 or .005, when using a single-night AHI, is likely to provide similar quantitative accuracy to detect a true positive association for a given outcome compared with multinight quantification of AHI/OSA severity with a conventional α cutoff. Ultimately, the choice of any significance threshold involves a trade-off between type I and II errors. However, one could argue that false discovery trials in medical science are more harmful than false nondiscovery trials. Thus, investigators may wish to consider variability in OSA severity when selecting an α significance level for a clinical trial. Furthermore, an α significance level of .005 has been suggested[38] and used in other fields[39,40] to increase the reproducibility of study findings by approximately 2-fold. Increasing the sample size and/or multinight recordings are likely to be the two possible solutions to reduce the type II errors. This suggests that existing epidemiologic and treatment trials to date may have been underpowered. In addition, high night-to-night variability in AHI may be an important and unique predictor of poor health outcomes in OSA.[11-13] Thus, it may be beneficial to consider using simple noninvasive monitoring of sleep apnea over multiple nights in addition to a more detailed polysomnographic assessment to better estimate OSA severity, its variability, and its impact. Whether night-to-night variability of other markers of OSA (eg, hypoxic burden,[41-43] sleep fragmentation,[44,45] and cardiovascular responses[46,47]) increases the uncertainty in detected associations with important health outcomes remains unknown and warrants further investigation.

Methodologic Considerations

The current dataset, analytical approach, and simulations allowed us to provide novel insights into OSA quantification and its impact on incident hypertension detection. Nonetheless, there are several limitations associated with the study design, including the lack of more refined assessment of OSA (hypoxemia, sleep fragmentation), together with a potentially greater AHI variability calculated with the WSA compared with polysomnography, that need to be acknowledged. Furthermore, the potential effects of

comorbid insomnia, medications, and other clinical covariates on device accuracy remain to be investigated. Despite these potential unknown influences, previous independent validation of the WSA showed good performance vs polysomnography-derived AHI.[1] Furthermore, AHI variability using the WSA is similar to that from multinight polysomnography studies.[1] The overall prevalence of OSA estimated with the WSA is also similar to population studies conducted with polysomnography.[1] The association between WSA-derived AHI with BP is also in accordance with existing epidemiologic trials.[13] The sample population consisted predominantly of self-selected males and is therefore not representative of the general population. The clinical presentation, manifestations, and pathophysiology of OSA, and response to treatment, differ in females compared with males.[48] Therefore, this study does not resolve the long-standing issue in sleep research of underrepresentation of females in such analyses, or the underlying assumption that thresholds derived from studies primarily of males will be valid or generalizable to women. Clinical covariates such as sleepiness, comorbid insomnia, and other OSA symptoms known to be predictive of poor health outcomes with OSA[49,50] also were not available in this study.

Information on hypertensive medications was not available in this study, and the follow-up time for incident hypertension was relatively short. Similarly, although participants received instructions on how to perform BP measurements, time of measurement was not standardized. However, BP variability observed in this study was low (approximately 7%) and similar to estimates reported in the literature.[51] In addition, height and weight used to calculate BMI were self-reported. These potentially confounding factors may have influenced the absolute magnitude of the reported hypertension associations. However, given that the outcome (hypertension) was fixed across all simulations, the reported increase in precision and magnitude of the association between OSA severity with multiple- vs single-night quantification and hypertension cannot be attributed to these potential confounding factors. Furthermore, as BP was calculated as the mean across multiple repeat measures during baseline and follow-up periods in the real-world setting, the reported BP values may be more informative than much of the previous OSA and hypertension literature, which has relied primarily on single time point assessments typically captured in the clinical setting.

Furthermore, information on potential OSA treatments was not available in this study. Some participants may also have sought treatment for OSA between baseline and follow-up. Thus, the effect size of the association between OSA severity and incident hypertension may have been underestimated. However, the reported increase in precision and magnitude of the association between OSA severity with multiple- vs single-night quantification was observed for all OSA severity categories; thus OSA treatment is unlikely to have confounded these results, at the very least in the moderate and severe OSA category.

Our study design required a high number of simulations (1,000), which would not have been possible using consecutive nights of AHI measurement. Thus, this approach differs from real-world trials, in which the variation over the first period of consecutive nightly monitoring may differ vs random sampling over a 3-month period. However, if anything, a random sampling approach may underestimate variability vs consecutive monitoring. For example, random sampling will minimize weekday vs weekend variation effects. Finally, the mattress sensor used in this study calculates the AHI automatically, based on physiological signals. Thus, our estimate of certainty does not incorporate the potential effect of interscorer variability, which is known to be significant with standard manually scored polysomnography.[52,53]

Interpretation

In summary, these data show that night-to-night variability in OSA severity is a major source of variance that negatively impacts the ability to accurately detect and draw inferences regarding the strength of associations between OSA severity and adverse health outcomes such as hypertension. These impacts are also likely to extend to treatment intervention trials as well as those that test for causal relationships. Multinight monitoring allows for more precise estimates of OSA severity and accurate detection of incident hypertension risk associated with OSA.

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Author contributions: B. L. is the guarantor of the content of the manuscript, including the data and analysis, and takes responsibility for the integrity of the work as a whole from inception to published article. B. L., P. G. C., and D. J. E. developed the study concepts. B. L., A. R., and K. L. performed the data extraction. B. L., D. P. N., and A. R. performed the data analysis. B. L., P. G. C., D. P. N., and D. J. E. drafted the manuscript. All authors provided important insight on data interpretation and contributed to the final version of the manuscript.

Role of sponsors: The sponsors had no role in the design of the study, the collection and analysis of the data, or the preparation of the manuscript.

Data sharing: Deidentified data that support the findings of this study, including individual data, are available from the corresponding author on request subject to ethical and data custodian (Withings) approval.

Additional information: The e-Figures are available online under "Supplementary Data."

References

1. Lechat B, Naik G, Reynolds A, et al. Multinight prevalence, variability, and diagnostic misclassification of obstructive sleep apnea. *Am J Respir Crit Care Med*. 2022;205(5):563-569.
2. Lang CJ, Appleton SL, Vakulin A, et al. Associations of undiagnosed obstructive sleep apnea and excessive daytime sleepiness with depression: an Australian population study. *J Clin Sleep Med*. 2017;13(4):575-582.
3. Nieto FJ, Peppard PE, Young T, Finn L, Hla KM, Farre R. Sleep-disordered breathing and cancer mortality: results from the Wisconsin Sleep Cohort Study. *Am J Respir Crit Care Med*. 2012;186(2):190-194.
4. Peppard PE, Szklo-Coxe M, Hla KM, Young T. Longitudinal association of sleep-related breathing disorder and depression. *Arch Intern Med*. 2006;166(16):1709-1715.
5. Peppard PE, Young T, Palta M, Skatrud J. Prospective study of the association between sleep-disordered breathing and hypertension. *N Engl J Med*. 2000;342(19):1378-1384.
6. Kendzerska T, Gershon AS, Hawker G, Leung RS, Tomlinson G. Obstructive sleep apnea and risk of cardiovascular events and all-cause mortality: a decade-long historical cohort study. *PLoS Med*. 2014;11(2):e1001599.
7. Young T, Peppard P, Palta M, et al. Population-based study of sleep-disordered breathing as a risk factor for hypertension. *Arch Internal Med*. 1997;157(15):1746-1752.
8. Punjabi NM, Caffo BS, Goodwin JL, et al. Sleep-disordered breathing and mortality: a prospective cohort study. *PLoS Med*. 2009;6(8):e1000132.
9. Punjabi NM, Patil S, Crainiceanu C, Aurora RN. Variability and misclassification of sleep apnea severity based on multi-night testing. *Chest*. 2020;158(1):365-373.
10. Roeder M, Bradicich M, Schwarz EI, et al. Night-to-night variability of respiratory events in obstructive sleep apnoea: a systematic review and meta-analysis. *Thorax*. 2020;75(12):1095-1102.
11. Linz D, Baumert M, Desteghe L, et al. Nightly sleep apnea severity in patients with atrial fibrillation: potential applications of long-term sleep apnea monitoring. *Int J Cardiol Heart Vasc*. 2019;24:100424.
12. Linz D, Brooks AG, Elliott AD, et al. Variability of sleep apnea severity and risk of atrial fibrillation: the VARIOSA-AF Study. *JACC Clin Electrophysiol*. 2019;5(6):692-701.
13. Lechat B, Loffler K, Reynolds A, et al. Association between night-to-night variability in obstructive sleep apnea severity with hypertension risk and blood pressure variability [abstract]. *Am J Respir Crit Care Med*. 2022;205:A3653.
14. Lechat B, Catcheside P, Reynolds A, Adams RJ, McEvoy RD, Eckert DJ. Reply to: Abreu and Punjabi, and Martinez-Garcia et al. *Am J Respir Crit Care Med*. 2022;206(1):126-129.
15. Simonds AK. How many more nights? Diagnosing and classifying obstructive sleep apnea using multinight home studies. *Am J Respir Crit Care Med*. 2022;205(5):491-492.
16. Martinez-Garcia MA, Oscullo G, Gomez-Olivas JD. Clinical implications of obstructive sleep apnea diagnostic misclassification. *Am J Respir Crit Care Med*. 2022;206(1):124-125.
17. Abreu A, Punjabi NM. How many nights are really needed to diagnose obstructive sleep apnea? *Am J Respir Crit Care Med*. 2022;206(1):125-126.
18. Heinzer R, Vat S, Marques-Vidal P, et al. Prevalence of sleep-disordered breathing in the general population: the HypnoLaus study. *Lancet Respir Med*. 2015;3(4):310-318.
19. Marshall NS, Wong KKH, Cullen SRJ, Knuiman MW, Grunstein RR. Sleep apnea and 20-year follow-up for all-cause mortality, stroke, and cancer incidence and mortality in the Busselton Health Study cohort. *J Clin Sleep Med*. 2014;10(04):355-362.
20. McEvoy RD, Antic NA, Heeley E, et al. CPAP for prevention of cardiovascular events in obstructive sleep apnea. *N Engl J Med*. 2016;375(10):919-931.
21. Drager LF, McEvoy RD, Barbe F, Lorenzi-Filho G, Redline S. INCOSACT Initiative (International Collaboration of Sleep Apnea Cardiovascular Trialists). Sleep apnea and cardiovascular disease: lessons from recent trials and need for team science. *Circulation*. 2017;136(19):1840-1850.
22. Sánchez-de-la-Torre M, Sánchez-de-la-Torre A, Bertran S, et al. Effect of obstructive sleep apnoea and its treatment with continuous positive airway pressure on the prevalence of cardiovascular events in patients with acute coronary syndrome (ISAACC study): a randomised controlled trial. *Lancet Respir Med*. 2020;8(4):359-367.
23. Barbe F, Duran-Cantolla J, Sanchez-de-la-Torre M, et al. Effect of continuous positive airway pressure on the incidence of hypertension and cardiovascular events in nonsleepy patients with obstructive sleep apnea: a randomized controlled trial. *JAMA*. 2012;307(20):2161-2168.
24. Nieto FJ, Young TB, Lind BK, et al. Association of sleep-disordered breathing, sleep apnea, and hypertension in a large community-based study: Sleep Heart Health Study. *JAMA*. 2000;283(14):1829-1836.
25. Cano-Pumarega I, Duran-Cantolla J, Aizpuru F, et al. Obstructive sleep apnea and systemic hypertension: longitudinal study in the general population: the Vitoria Sleep Cohort. *Am J Respir Crit Care Med*. 2011;184(11):1299-1304.
26. O'Connor GT, Caffo B, Newman AB, et al. Prospective study of sleep-disordered breathing and hypertension: the Sleep

Heart Health Study. *Am J Respir Crit Care Med*. 2009;179(12):1159-1164.

27. Bratton DJ, Stradling JR, Barbe F, Kohler M. Effect of CPAP on blood pressure in patients with minimally symptomatic obstructive sleep apnoea: a meta-analysis using individual patient data from four randomised controlled trials. *Thorax*. 2014;69(12):1128-1135.
28. Ponce S, Pastor E, Orosa B, et al. Sleep Respiratory Disorders Group of the Sociedad Valenciana de Neumología. The role of CPAP treatment in elderly patients with moderate obstructive sleep apnoea: a multicentre randomised controlled trial. *Eur Respir J*. 2019;54(2):1900518.
29. Pengo MF, Soranna D, Giontella A, et al. Obstructive sleep apnoea treatment and blood pressure: which phenotypes predict a response? A systematic review and meta-analysis. *Eur Respir J*. 2020;55(5):1901945.
30. Topouchian J, Zelveian P, Hakobyan Z, Gharibyan H, Asmar R. Accuracy of the Withings BPM connect device for self-blood pressure measurements in general population - validation according to the Association for the Advancement of Medical Instrumentation/European Society of Hypertension/International Organization for Standardization Universal Standard. *Vasc Health Risk Manag*. 2022;18:191-200.
31. Edouard P, Campo D, Bartet P, et al. Validation of the Withings Sleep Analyzer, an under-the-mattress device for the detection of moderate-severe sleep apnea syndrome. *J Clin Sleep Med*. 2021;17(6):1217-1227.
32. Epstein LJ, Kristo D, Strollo PJ Jr, et al. Clinical guideline for the evaluation, management and long-term care of obstructive sleep apnea in adults. *J Clin Sleep Med*. 2009;5(3):263-276.
33. Chobanian AV, Bakris GL, Black HR, et al. The Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure: the JNC 7 report. *JAMA*. 2003;289(19):2560-2572.
34. Niiranen TJ, Hanninen MR, Johansson J, Reunanen A, Jula AM. Home-measured blood pressure is a stronger predictor of cardiovascular risk than office blood pressure: the Finn-Home study. *Hypertension*. 2010;55(6):1346-1351.
35. Ward AM, Takahashi O, Stevens R, Heneghan C. Home measurement of blood pressure and cardiovascular disease: systematic review and meta-analysis of prospective studies. *J Hypertens*. 2012;30(3):449-456.
36. Stergiou GS, Asayama K, Thijs L, et al. Prognosis of white-coat and masked hypertension: International Database of HOme blood pressure in relation to Cardiovascular Outcome. *Hypertension*. 2014;63(4):675-682.
37. Stergiou GS, Palatini P, Parati G, et al. 2021 European Society of Hypertension practice guidelines for office and out-of-office blood pressure measurement. *J Hypertens*. 2021;39(7):1293-1302.
38. Benjamin DJ, Berger JO, Johannesson M, et al. Redefine statistical significance. *Nat Hum Behav*. 2018;2(1):6-10.
39. Open Science Collaboration. PSYCHOLOGY. Estimating the reproducibility of psychological science. *Science*. 2015;349(6251):aac4716.
40. Nosek BA, Hardwicke TE, Moshontz H, et al. Replicability, robustness, and reproducibility in psychological science. *Annu Rev Psychol*. 2022;73:719-748.
41. Trzepizur W, Blanchard M, Ganem T, et al. Sleep apnea-specific hypoxic burden, symptom subtypes, and risk of cardiovascular events and all-cause mortality. *Am J Respir Crit Care Med*. 2022;205(1):108-117.
42. Azarbarzin A, Sands SA, Stone KL, et al. The hypoxic burden of sleep apnoea predicts cardiovascular disease-related mortality: the Osteoporotic Fractures in Men Study and the Sleep Heart Health Study. *Eur Heart J*. 2019;40(14):1149-1157.
43. Azarbarzin A, Sands SA, Taranto-Montemurro L, et al. The sleep apnea-specific hypoxic burden predicts incident heart failure. *Chest*. 2020;158(2):739-750.
44. Lechat B, Hansen KL, Melaku YA, et al. A novel electroencephalogram-derived measure of disrupted delta wave activity during sleep predicts all-cause mortality risk. *Ann Am Thorac Soc*. 2022;19(4):649-658.
45. Lechat B, Scott H, Naik G, et al. New and emerging approaches to better define sleep disruption and its consequences. *Front Neurosci*. 2021;15:751730.
46. Azarbarzin A, Sands SA, Younes M, et al. The sleep apnea-specific pulse rate response predicts cardiovascular morbidity and mortality. *Am J Respir Crit Care Med*. 2021;203(12):1546-1555.
47. Kwon Y, Wiles C, Parker BE, et al. Pulse arrival time, a novel sleep cardiovascular marker: the Multi-Ethnic Study of Atherosclerosis. *Thorax*. 2021;76(11):1124-1130.
48. Won CHJ, Reid M, Sofer T, et al. Sex differences in obstructive sleep apnea phenotypes, the Multi-Ethnic Study of Atherosclerosis. *Sleep*. 2020;43(5):zsz274.
49. Lechat B, Appleton S, Melaku YA, et al. Comorbid insomnia and obstructive sleep apnoea is associated with all-cause mortality. *Eur Respir J*. 2022;60(1):2101958.
50. Mazzotti DR, Keenan BT, Lim DC, Gottlieb DJ, Kim J, Pack AI. Symptom subtypes of obstructive sleep apnea predict incidence of cardiovascular outcomes. *Am J Respir Crit Care Med*. 2019;200(4):493-506.
51. Mehlum MH, Liestol K, Kjeldsen SE, et al. Blood pressure variability and risk of cardiovascular events and death in patients with hypertension and different baseline risks. *Eur Heart J*. 2018;39(24):2243-2251.
52. Kuna ST, Benca R, Kushida CA, et al. Agreement in computer-assisted manual scoring of polysomnograms across sleep centers. *Sleep*. 2013;36(4):583-589.
53. Ruehland WR, Rochford PD, O'Donoghue FJ, Pierce RJ, Singh P, Thornton AT. The new AASM criteria for scoring hypopneas: impact on the apnea hypopnea index. *Sleep*. 2009;32(2):150-157.